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Effects of audio-visual interactions on soundscape and landscape perception and their influence on satisfaction with the urban environment

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ABSTRACT

This study investigated the relationship between overall satisfaction with the urban environment and its soundscape and landscape based on a subjective evaluation of the audio-visual interaction effects within various urban spaces with different environmental conditions. Actual site conditions were simulated using immersive virtual reality technology in which subjects were provided with visual information via a head-mounted display (HMD) and audio information via head-tracking technology using the first-order ambisonics (FOA) of headphone-based three-dimensional auralization. Subjective responses to sound sources and visual elements such as cognitive response, semantic expression, and rated satisfaction were investigated in audio-visual as well as audio- and visual-only environments. It was shown that the availability of visual information affects the auditory perception of a number of human-made and natural sounds and the availability of audio information affects the visual perception of various visual elements. Semantic differential evaluation revealed that *pleasantness* and *eventfulness* and *overall quality*, *regularity*, *spatial impression*, and *naturalness* were the main evaluation factors for soundscapes and landscapes, respectively. One new finding was that audio information affects the perception of the naturalness of a landscape. Audio and visual information had effects of 24 and 76%, respectively, on overall satisfaction. Based on our findings, overall satisfaction models based on audio-visual elements and soundscape and landscape components, which exhibited 31 and 51% explanatory powers, respectively, were proposed. These newly proposed audio-visual interaction-based satisfaction models can be used for better urban planning.

1. Introduction

Noise pollution in urban spaces is rapidly being recognized as a social issue with impacts on human health [1–3]. To reflect the vital importance of a pleasant urban acoustic environment to a sustainable, healthy city, the concept of soundscape, which concerns an individual's perception of their acoustic environment, has been developed [4–6]. ISO 12913-1 defines soundscape as "the acoustic environment as perceived or experienced and/or understood by a person or people, in context" [7]. Many studies have examined the correlations between acoustic indicators and soundscape descriptors (such as comfort, tranquility, and satisfaction) to evaluate the perception of environmental noise in spaces such as parks, squares, and streets [8–12]. Because context is known to influence soundscape perception through both on-site non-acoustic and acoustic factors, the correlations between soundscape and context have also been examined [13,14].

Because the primary soundscape concept relates to landscape and 80% of human sensory experiences depend on visual stimuli [15], the correlations between landscape and soundscape have been investigated extensively [16–19]. On a larger scale, the correlations between landscape spatial patterns (e.g., urban morphology) and the soundscapes of entire cities have also been investigated [20–23]. Most studies on soundscape auditory reactions have focused on landscape auditory properties; however, despite the fact that human perception has a large dependence on visual factors, most landscape studies on auditory reaction to soundscapes have been restricted to the analysis of auditory properties. Thus, an in-depth understanding of the overall perception of the urban environment requires a consideration of the visual reaction to the landscape. Although the effects of audio-visual or soundscape-landscape interaction on soundscape perception have also been examined [24], such studies have been methodologically limited by their reliance on the evaluation of visual stimuli using conventional two-dimensional photographs, collages, or Photoshop images that do not fully reflect an actual on-site sense of space or reality [8,25–28].

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Furthermore, most studies have used objective metrics, such as the height, area, and/or circumference of visual elements (e.g., vehicles, buildings, and green areas), to evaluate perceived visual effects and their correlations with the perception of soundscape. As the soundscape pertains to actual human perceptual reaction, which is largely subjective, landscape evaluation also requires human-reaction-based analysis to account for subjective perception.

Many studies on the human perception of landscape have been conducted [29,30]. In this context, landscape is defined as "an area, as perceived by people, whose character is the result of the action and interaction of natural and/or human factors" (Council of Europe 2000) and is tightly connected to human perception. Landscape evaluation can be primarily divided into landscape preference and visual aesthetic quality [29,31], both of which have been reported to be affected by meaning and sense of place (memories, historical/cultural/social significance, ethical/moral obligation, and spiritual values) [32,33]. In addition, visual elements in the design aspect of the landscape [34] and the significant effects of space, well-preserved human-derived elements, and vegetation proportion on preferences with respect to visual quality [35] have been studied. Interest in the effects of sound on landscape has also increased, resulting in studies on the effect of sound on landscape value [36], which have shown that congruence or coherence between the sound with the landscape significantly affects preference. Further investigation of the effects of sound type on the evaluation of natural landscapes have revealed that anthropogenic noise (air and ground traffic, voices, etc.) have a negative effect while natural sounds (bird calls, breeze through foliage, etc.) have a positive effect [37]. Despite the number of approaches taken to understanding landscape perception, most sound-based studies have been based on the use of 2D images. To gain a better understanding of the relationship between sound and landscape perception, it will necessary to conduct research in environments that are more ecologically realistic.

In this context, virtual reality (VR) technology can create immersive environments that emulate actual urban spaces under controlled laboratory conditions. VR has been applied to soundscape evaluation [38–40] and Studies on the validity of VR technology include work by Luigi et al. [41], who compared global, acoustic, and visual qualities of the real world with those of the same environments reproduced using a VR tool and found no significant difference in the subjective evaluation results. Puyana-Romero et al. [42] extended the validation of VR technology to the evaluation of the urban environment evaluation using a head-mounted display (HMD), mobile-based HMD, and laptop or desktop computer and also found no significant difference in subjective evaluation results between the real and VR environments. The results of [41,42] suggest that subjective evaluation using VR technology can produce sufficiently valid results. Hong et al. [43] evaluated headphone- and speaker-based VR environments and concluded that a first-order ambisonic (FOA) reproduction method reflecting headphone-based head rotation yielded effects similar to those obtained at an actual site in terms of overall soundscape quality and sound- and environment-related spatial attributes. Lindquist et al. [44] found that proper congruence of vision and sound in a VR-modeled environment can lead to very high degrees of realism. Overall, these results suggest the applicability of VR technology to progressive, practical research on audio-visual or soundscape−landscape interactions.

Although a number of models of audio-visual interaction for urban environment (UE) interpretation have been proposed [45–47], to date few urban-soundscape-perception interpretation models have been developed. Hong & Jeon [48] proposed a structural equation-based model for various noise sources, soundscape perceptions, visual attitudes, and aural/visual environment harmonies. In addition, several analytic models of acoustic comfort, environmental acceptability, and preference have been developed [49], although none has provided sufficient acoustic analytical capability or the capacity to consider landscape-related factors.

To address these gaps in the research literature, we investigated the effects of audio-visual interaction on soundscape and landscape perception in a UE using VR and examined how the results correlated with overall UE satisfaction on the part of human subjects. We also performed an auditory evaluation of different evaluation environments (EEs) based on available aural and visual information.

2. Methodology

2.1. Audio and visual materials

2.1.1. Case study area

To investigate the audio-visual factors influencing soundscape and landscape perception through auditory evaluation in a laboratory environment, we selected nine evaluation locations (ELs) across Seoul representing various urban contexts. Images of these places, captured from location videos, are shown in Fig. 1. Proceeding from the noisiest location (as a result of road proximity and traffic congestion) to a relatively tranquil location surrounded by buildings, we chose locales, including open land, green areas (vegetation), and rivers, in which there were a variety of noise sources, such as traffic, people, and birds.

2.1.2. Audio-visual recording

To construct the laboratory VR EEs, we performed on-site measurements at daytime (10 a.m.–2 pm) during September 2018. At each location, we recorded omnidirectional videos at 8 k ultra-high definition, 30-fps resolution, and 95 Mbps for 3 min using a spherical panoramic camera (Insta 360 camera). To obtain audio-environment data, we used an ambisonic microphone (Sound field SPS 200) and a portable sound recorder (Mixpre-6, Sound devices), with A-format FOA recording [41,50,51]. The ambisonic microphone had a flat frequency response in the range 04 Hz–20 kHz, a sensitivity of 14 mV/Pa, and a maximum sound pressure level (SPL) with a performance of 130 dB. The portable sound recorder performed at ±0.5 dB in the range 10 Hz–80 kHz and a total harmonic distortion within 0.005%, indicating a high degree of accuracy. With the spherical camera and ambisonic microphone positioned at a height of 1.6 m, we measured A-weighted SPLs (L_{Aeq}) using a 1/2-inch microphone (G.R.A.S AE46) and an AS-70 portable sound-level meter (RION) to calibrate the SPL for laboratory implementation of the virtual soundscape EE. The 1/2-inch microphone had a flat frequency response at 3.15 Hz–20 kHz, a sensitivity of 50 mV/Pa, a dynamic range of 17–138 dB, and, as measured by a portable sound-level meter, an accuracy of ±1 dB in the range 0.25 Hz–20 kHz. We used a clapping sound to synchronize the measured video and audio data prior to measurement [43].

2.1.3. VR soundscape environment reproduction

We interconnected the six-channel 360-degree video to an HMD via post-processing (Insta360 stitcher). To produce headphone-based stereo sound, we converted the A-format FOA recording to B-format FOA using Unity software and down-mixed to binaural tracks. To improve the sense of reality in the audio-environment, we implemented real-time subject-head-rotation direction adjustment using a VR-mounted head tracker. We achieved video and sound synchronization using clapping.

To calibrate the SPL at each EL, we recorded nine VR-implemented sound sources using a head-and-torso simulator (Brüel & Kjær 4100) placed in a semi-anechoic chamber. Sound pressure calibration at the actual measurement site was achieved by calculating the A-weighted equivalent continuous sound level ($L_{\mathrm{Aeq,3\text{-}min}}$) of each sound source and adjusting it using Adobe Audition (version 1.5) to equalize it with the $L_{\mathrm{Aeq,3\text{-}min}}$ of the microphone-recorded sound source [38,43,52,53].

2.1.4. Acoustic indicators

To investigate the acoustic environmental characteristics of each EL, we analyzed indicators relating to sound strength, spectral content, and temporal structure (Appendix A). To analyze the sound strength characteristics, we calculated percentile SPLs, including L_{Aeq}, L_{A10}, L_{A50}, and

Fig. 1. Panoramic views of nine ELs used in experiment.

L_{A90}. To obtain sound spectral content, we analyzed $L_{Ceq\text{-}Aeq}$, which represents relatively low-frequency characteristics and constitutes the difference between the A- and C-weighted SPLs. The average L_{Aeq} values of the ELs spanned a wide range from 57.2 to 79.4 dBA, and the differences between the maximum and minimum $L_{A10\text{-}A90}$ and $L_{Ceq\text{-}Aeq}$ exceeded 10 dB, encompassing the entire variation range of the acoustic environment that can occur in a general urban space.

2.2. Experiment design

2.2.1. Participants

Thirty subjects participated in the study, a number larger than the previously suggested proper sample size for reliable investigation of audio-visual stimuli in a laboratory-implemented VR EE [8,16]. To reduce intra-group variation, we chose students from our own university (Hanyang University) as subjects (Male = 20, Female = 10). The subjects ranged in age from 22–29 years (mean age: 24.80 years, standard deviation: 1.97 years). Prior to evaluation, we assessed the subjects' audible limits using an audiometer (Rion AA-77); all were found to have normal hearing.

2.2.2. Experimental setups

The main objective was to examine the effects of audio-visual interaction on urban audio-visual environment perception. We evaluated the soundscape in three EEs: (1) audio-only, (2) visual-only, and (3) audio-visual. We performed the experiment in a semi-anechoic chamber with ~25-dBA background noise. We provided aural and visual information via open-type headphones (Sennheiser HD-650) and a VIVE Pro 2 HMD, respectively.

2.2.3. Questionnaire

The evaluation questionnaire for audio-visual interaction with the soundscape was administered in five parts, as detailed in Appendix B. Question 1 investigated the prominent sound source at each EL. Axelsson, Nilsson, & Berglund [54] proposed a five-category classification system (traffic, fans, other, human-derived, and natural sounds) as a questionnaire protocol for soundscape quality evaluation. Along with water (which was excluded from the study target locations), the five sound categories used in this study (Appendix B, Q1) are the most frequently used in soundscape evaluation research [23,45,48].

The visual environment is closely correlated with sound environment perception, with the visual environmental quality strongly influencing the soundscape quality [19,55,56]. Accordingly, the second question concerned the prominent visual elements at each EL. The visual elements considered in this question (Appendix B, Q2) had previously been used to objectively evaluate landscapes [8,23,46,57].

A number of researchers have investigated soundscape perception through soundscape factors based on the use of verbal descriptors or principal component analysis (PCA) [9,23,48,58–60]. Applying this methodology, question 3 was used to evaluate the acoustic environment perception for each EL using the eight semantic attributes detailed in Appendix B (Q3) [60,61]. Based on previous findings [62–65] that landscape influences soundscape perception, question 4 used 12 semantic attributes (Appendix B, Q4) to uncover the influence of landscape in each EL. Question 5 was used to evaluate the overall satisfaction at each EL.

2.3. Procedure

Each subject evaluated the soundscape in audio-only, visual-only, and audio-visual environments in a random order to exclude any ordering effect, experiencing one EE per day for three days. The questionnaire format was changed according to the EE. For the audio- and visual-only cases, questions 1, 3, and 5 and 2, 4, and 5, respectively, were posed. For the audio-visual case, all questions were posed. We used the Unity VR platform to enable the subjects to select answers directly in the VR environment through a VIVE controller. The subjects followed the recording path in sequentially evaluating nine stimuli (H1 to H9, Appendix A) with a 10-s interval between each sound source (each sound was 180-s long). To relieve physical discomfort of prolonged HMD use, we imposed a 30-min experimental time for each EE, with sufficient rest time provided upon the subject's request.

2.4. Data analysis

To conduct statistical analysis, we used SPSS v.25 to perform normality (Shapiro−Wilk) and homoscedasticity (Levene) testing to verify the parametric statistic applicability of all answers. As the same group participated in the experiment under all three different experimental environments, a two-way repeated measure analysis of variance test (RM ANOVA) was performed to investigate the effect of audio and visual information on soundscape, while principal component analysis (PCA) was used to extract prominent factors for the soundscape- and landscape-related semantic attributes. The orthogonal components and number of components were determined via varimax rotation and by setting a criterion of an eigenvalue of one or higher, respectively. To examine the influence of the audio-visual elements on overall satisfaction, we determined the contribution of each factor using multiple linear regression (MLR).

3. Results

3.1. Audio-visual indicator perception

3.1.1. Perceived prominent sound sources

Fig. 2 shows the prominent sounds and individual sound preferences at each EL for the audio-only and audio-visual EEs. The response percentages indicate the ratio of subjects who responded "a lot" or

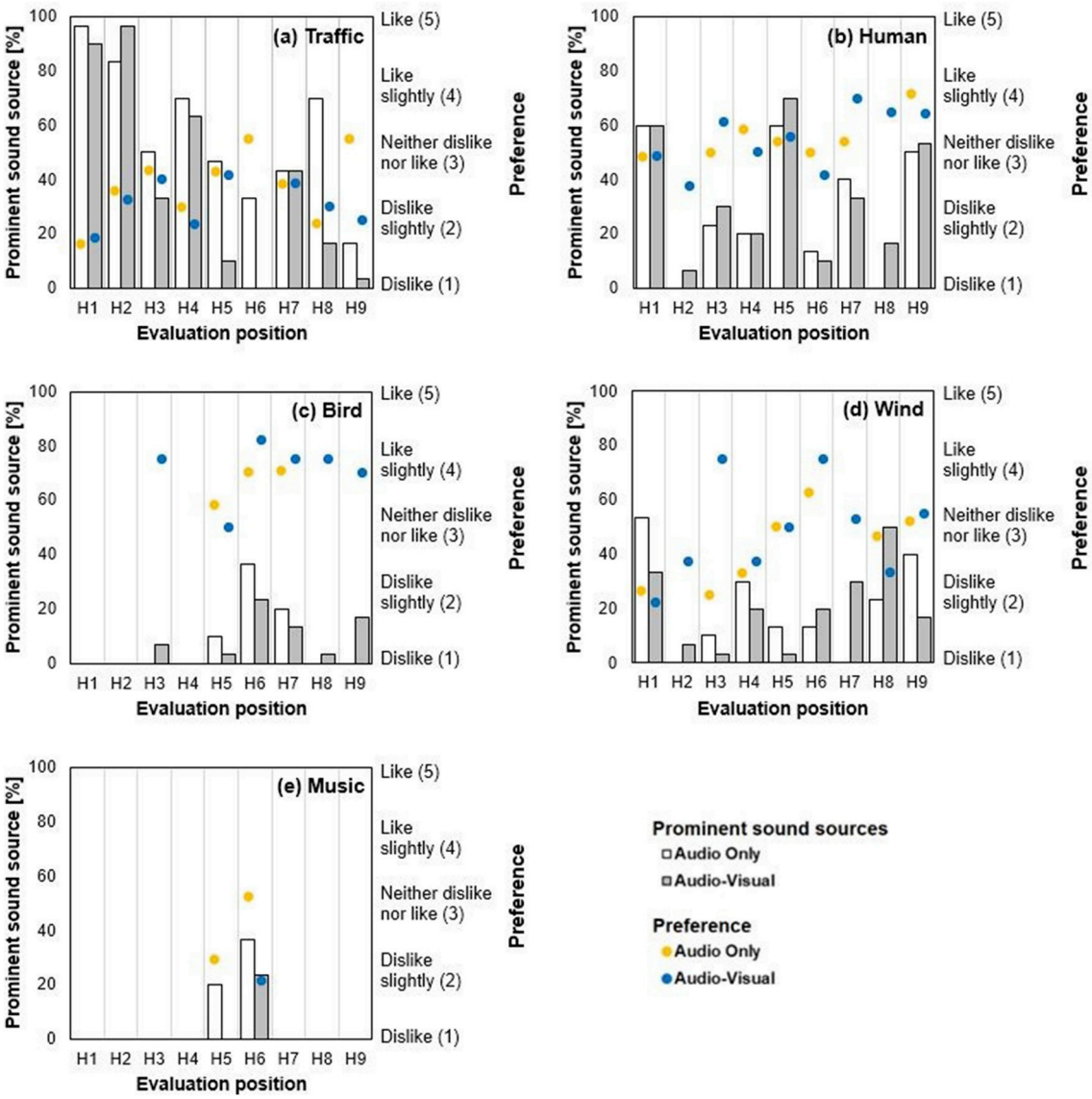

Fig. 2. Prominent sound sources and preferences for each location: (a) traffic, (b) human, (c) bird, (d) wind, and (e) music sounds.

"dominates" for each sound source. Traffic sounds in general were most prominent, followed by human, wind, bird, and music sounds. Traffic was the most prominent at locations closer to roads (H1, H2, H3, H4). At H1 and H2, which were the closest to a road, the perception rate exceeded 80% regardless of the EE. Up to 96.7% of the subjects perceived road traffic noise. The human-sound perception rate was high near bus stops/subway stations (H1), buildings (H7, H9), and main routes (H5). The highest perception rate was 70%, at H5. Bird sounds were perceived well at green-space locations with low traffic noise (H6, H7, H9); the highest bird sound perception rate (36.7%) was obtained on a walking trail (H6). Wind sounds were relatively high in locations with a large number of vehicles (H1) or in open spaces. Music was perceived from speakers near the EL (H5, H6).

Traffic sounds had the lowest mean preference value (audio: 2.59, audio–visual: 2.54), while bird (audio: 3.20, audio–visual: 3.14) and human sounds (audio: 3.16, audio–visual: 3.25) were positively evaluated. The differences in preferences among the EEs were not statistically significant, indicating similar preferences for each type of sound source irrespective of the visual information. We used the Pearson correlation to investigate the correlation between the extent to which a sound was heard and the subjects' preference for that sound in each EE. The Pearson correlations for "traffic" were −0.91 and −0.84 for the audio-only and audio-visual environments, respectively, showing significant negative correlations ($p < 0.01$), indicating that traffic sounds were evaluated more negatively in places where more traffic sounds were received by subjects. By contrast, a strong positive correlation of 0.91 was found in audio-visual environments ($p < 0.01$) containing bird sounds, indicating that the subjects more positively evaluated the sound of birds in places in which it was perceived more prevalently.

3.1.2. *Visual information effect on soundscape index perception*

As shown in Fig. 2, the sound-source perception rates differed between the audio-only and audio-visual environments. We applied a two-way repeated measures ANOVA model to assess the significance of the difference in the average sound-source perception reactions according to the visual information provided and location (Table 1). For visual information, the differences in the perception of traffic, human, bird, and wind (excluding music) sources were statistically significant ($p < 0.01$); with the exception of "wind," the effect sizes were all at least 0.3, which is greater than the average effect size. The "traffic" five-point scale scores were higher in the audio-only environments (mean: 3.52, SD: 1.2) than in the audio-visual environments (mean: 3.12, SD: 1.14). The average scores for human, bird, and wind sounds in the audio-visual environments exceeded those in the audio-only environments by 0.37,

Table 1
Main and interaction effects and simple effect of visual information and location for different sound sources.

Indicator	Factor	Sum of squares	Degree of freedom	Mean square	F	p-value	ES (η_p^2)
Main effect and interaction effect							
Traffic	Visual	22.00	1	22.00	16.06	<0.01	0.36
	Location	275.33	8	34.42	38.10	<0.01	0.57
	Interaction	28.15	8	3.52	6.50	<0.01	0.18
Human	Visual	18.89	1	18.89	12.25	<0.01	0.30
	Location	205.98	8	25.75	40.12	<0.01	0.58
	Interaction	13.03	8	1.63	2.95	<0.01	0.09
Bird	Visual	35.78	1	35.78	54.14	<0.01	0.65
	Location	95.19	8	11.90	16.43	<0.01	0.36
	Interaction	13.50	8	1.69	3.40	<0.01	0.11
Wind	Visual	15.00	1	15.00	7.33	<0.01	0.20
	Location	39.09	8	4.89	4.42	<0.01	0.13
	Interaction	29.83	8	3.73	5.22	<0.01	0.15
Music	Visual	0.12	1	0.12	0.07	0.80	0.00
	Location	60.77	8	7.60	14.28	<0.01	0.33
	Interaction	15.68	8	1.96	3.62	<0.01	0.11
Simple effect							
Traffic	H5	17.07	1	17.07	19.09	<0.01	0.40
	H6	14.02	1	14.02	18.92	<0.01	0.40
	H8	16.02	1	16.02	17.54	<0.01	0.38
Human	H2	8.07	1	8.07	16.79	<0.01	0.37
	H3	8.82	1	8.82	10.34	<0.01	0.26
	H8	11.27	1	11.27	6.90	<0.01	0.19
Bird	H3	14.02	1	14.02	32.56	<0.01	0.53
	H4	12.15	1	12.15	47.94	<0.01	0.62
	H8	7.35	1	7.35	16.21	<0.01	0.36
	H9	10.42	1	10.42	16.71	<0.01	0.37
Wind	H7	20.42	1	20.42	31.03	<0.01	0.52
	H8	12.15	1	12.15	11.61	<0.01	0.29

0.51, and 0.34, respectively.

At various ELs, mutual interactions were present between the provided sound and visual information. The statistically significant simple main effects of adding visual information are listed in Table 1. Significant differences in the perception of traffic sound were found at H5, H6, and H8, with focus on traffic sound decreasing with the provision of visual information. Similarly, significant differences for human, bird, and wind sound perception were found at H2, H3, and H8, at H3, H4, and H8, and at H7 and H8, respectively. When visual information was supplied, the dominance of traffic sound decreased, reflecting the expected effects of visual elements at the location [66]. Thus, although some location dependence was observed, the significant influence of visual information on the sound sources comprising a soundscape was confirmed.

3.1.3. Perceived prominent visual elements

Fig. 3 shows the perceived prominent visual elements and individual visual-element preferences at each EL for the visual-only and audio-visual environments. The response percentages indicate the ratio of subjects who responded "a lot" or "dominates" with respect to visual element response. More than 50% perceived the "vehicle" and "road" elements at locations close to large roads (H1–H4) in both environments. Although "road" was perceived at some locations far from large roads (H7–H9), the visual perception of "traffic" in these cases was relatively low as a result of the low vehicle flow at the time of the evaluation combined with the physical distance from any large roads.

"Building" was perceived at every location except the walking trail (H6). The "open space" perception rate was very high, exceeding 80% near roadside parks (H2, H3). More than 50% of subjects perceived "open space" in large pedestrianized spaces (H4, H5) and around buildings (H8, H9). The "green space" perception rate was high at a park (H3) and on a trail (H6).The "people" perception rate was high at a public transportation platform located close to a large road (H1) and near a building (H8) and a main route (H5, which had the highest score, exceeding 80%).The visual perception for "sky" was higher for locations further from complex roads.

The lowest average preference was for "vehicle" (visual: 2.60, audio–visual: 2.78). The "road" preference was also low, with a value of 2.91 in the visual environment. The preferences for "open space," "vegetation," "people," and "sky" were all 3.43 or greater, indicating positive evaluations for these elements. Independent t-testing revealed statistically significant differences in preference according to EE for all visual elements except "vehicle" and "open space" (i.e., "building," "road," "vegetation," "people," and "sky") ($p < 0.01$). The Pearson correlation was used to determine the correlation between the visual element perception rate and preference for each EE. The correlations in the visual environment for "road," "open space," and "sky" were significant (−0.83, −0.72, and 0.71, respectively), suggesting that the subjects evaluated roads and open spaces more negatively when perceiving them more strongly but evaluated the sky more positively when it was more visible ($p < 0.05$).

3.1.4. Influence of audio information on landscape index perception

Fig. 3 indicates that the visual-element perception rates differed between the visual-only and audio-visual environments. To assess the significance of the average differences in the subjects' visual-element perception reactions according to audio information provision and EL (Table 2), a two-way repeated measures ANOVA model was applied. The perceptions of "vehicle," "building," "vegetation," "people," and "sky" elements differed significantly between the audio-visual and visual-only cases. Because there were correlations between the respective evaluation points and the availability of audio information for the "vehicle," "building," and "vegetation" categories, the simple main effect was analyzed for each variable. The analysis revealed statistically significant differences with high effect sizes at specific places for each of the categories, indicating that the effect of audio information is significant in evaluating satisfaction. The audio-visual environment had a higher perception value for "vehicle" (mean: 3.10, SD: 1.33) than the visual-

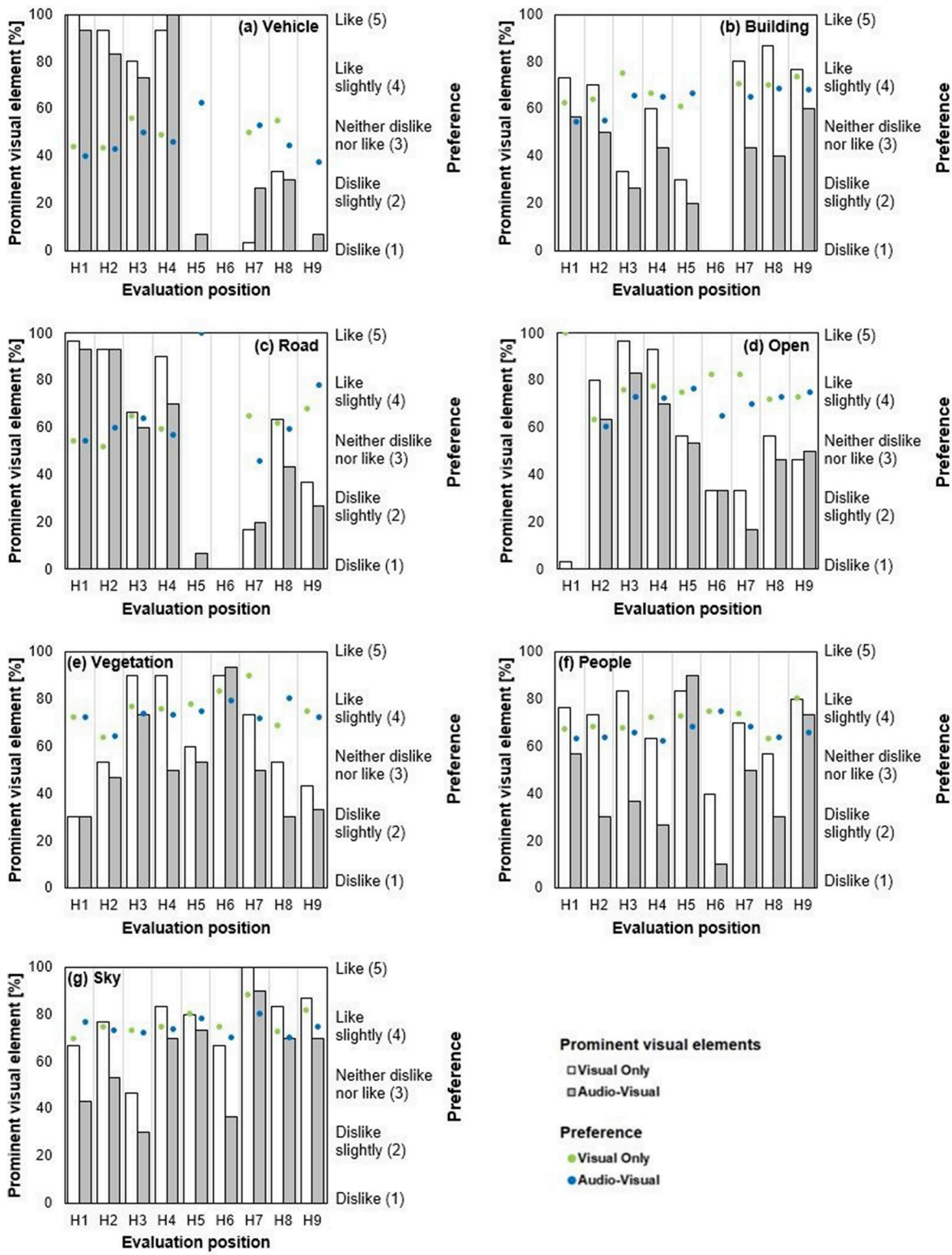

Fig. 3. Prominent visual elements and associated preferences at each location: (a) "vehicle," (b) "building," (c) "road," (d) "open," (e) "vegetation," (f) "people," and (g)" sky".

only environment (mean: 2.94, SD: 1.54), while the perception of "building" in the audio-visual environment was worse (mean: 3.33, SD: 0.74) than in the visual-only environment (mean: 3.61, SD: 0.86). "Vegetation" scored higher in the visual-only environment (mean: 3.75, SD: 0.91) than in the audio-visual environment (mean: 3.49, SD: 0.77), while the perceptions of "people" and "sky" were worse (by 0.41 and 0.34, respectively) in the visual-only environment. There were no significant differences in perception by EE for "road" or "open space," suggesting that elements that appear sufficiently prominent with visual-only information are less affected by additional audio information.

Table 2
Main effects and interaction effects and simple effect of audio information and location for different visual elements.

Indicator	Factor	Sum of squares	Degree of freedom	Mean square	F	p-value	ES (η_p^2)
Main effect and interaction effect							
Vehicle	Audio	3.75	1	3.75	4.28	<0.05	0.13
	Location	788.23	8	98.53	160.98	<0.01	0.85
	Interaction	16.30	8	2.04	4.11	<0.01	0.12
Building	Audio	10.98	1	10.98	14.95	<0.01	0.34
	Location	79.75	8	9.97	25.01	<0.01	0.46
	Interaction	6.10	8	0.76	2.39	<0.05	0.08
Road	Audio	0.05	1	0.05	0.08	0.80	0.01
	Location	560.68	8	70.09	76.86	<0.01	0.77
	Interaction	13.97	8	1.75	4.46	<0.01	0.13
Open	Audio	0.54	1	0.54	0.43	0.52	0.02
	Location	181.10	8	22.64	34.91	<0.01	0.55
	Interaction	14.02	8	1.75	4.20	<0.01	0.13
Vegetation	Audio	9.34	1	9.34	11.00	<0.01	0.28
	Location	79.20	8	9.90	19.91	<0.01	0.41
	Interaction	5.55	8	0.69	2.09	<0.01	0.07
People	Audio	22.82	1	22.82	21.64	<0.01	0.43
	Location	53.28	8	6.66	19.22	<0.01	0.40
	Interaction	3.40	8	0.43	1.74	0.09	0.06
Sky	Audio	17.42	1	17.42	27.95	<0.01	0.49
	Location	60.26	8	7.53	17.46	<0.01	0.38
	Interaction	3.19	8	0.40	1.66	0.11	0.05
Simple effect							
Vehicle	H1	1.35	1	1.35	4.28	<0.05	0.13
	H5	3.75	1	3.75	5.51	<0.05	0.16
	H6	2.40	1	2.40	19.33	<0.01	0.40
	H7	3.75	1	3.75	5.8	<0.05	0.17
Building	H7	3.27	1	3.27	9.73	<0.01	0.25
	H8	7.35	1	7.35	29.81	<0.01	0.51
	H9	2.02	1	2.02	7.82	<0.01	0.21
Vegetation	H3	2.02	1	2.02	10.67	<0.01	0.27
	H4	4.27	1	4.27	26.14	<0.01	0.47
	H7	5.40	1	5.40	20.61	<0.01	0.42

Table 3
PCA-rotated component matrices for semantic expressions concerning soundscape and landscape (numbers in parentheses: explained variance).

Set-ups	Audio Only		Audio-visual	
Soundscape component	Pleasantness (33.67)	Eventfulness (32.75)	Eventfulness (36.94)	Pleasantness (30.99)
Pleasant	**0.78**	0.31	0.25	**0.84**
Unpleasant	**−0.82**	−0.17	−0.02	**−0.84**
Calm	**0.76**	−0.19	−0.38	**0.62**
Chaotic	**−0.85**	0.10	0.28	**−0.71**
Eventful	0.11	**0.71**	**0.86**	−0.17
Uneventful	0.07	**−0.78**	**−0.82**	0.20
Exciting	0.30	**−0.19**	**0.80**	0.32
Monotonous	0.16	**0.10**	**−0.79**	0.09

Landscape component	Visual Only			Audio-visual			
	Overall quality (27.66)	Spatial impression (19.02)	Regularity (15.69)	Overall quality (20.33)	Regularity (17.41)	Spatial impression (17.30)	Naturalness (13.40)
Comfortable	**0.83**	0.14	0.17	**0.62**	0.38	0.05	0.35
Uncomfortable	**−0.73**	−0.16	−0.25	**−0.61**	−0.49	−0.08	−0.20
Harmonious	**0.71**	0.13	0.31	**0.87**	0.07	0.09	0.13
Disharmonious	**−0.64**	−0.01	−0.42	**−0.87**	−0.10	−0.04	−0.03
Natural	**0.80**	−0.03	0.01	0.24	0.31	0.02	**0.76**
Artificial	**−0.69**	0.09	0.21	−0.10	0.08	0.05	**−0.89**
Open	0.08	**0.74**	0.13	−0.01	0.34	**0.67**	0.13
Closed	−0.02	−0.39	**0.52**	0.07	0.43	**−0.53**	−0.09
Wide	0.08	**0.87**	0.09	0.16	0.07	**0.78**	−0.04
Narrow	0.01	**−0.83**	−0.02	−0.07	−0.12	**−0.81**	0.13
Orderly	0.16	0.20	**0.71**	0.18	**0.78**	0.12	0.11
Disorderly	−0.19	−0.19	**−0.83**	−0.20	**−0.81**	−0.20	−0.05

3.2. Semantic differential test

3.2.1. Soundscape principal components

We performed a PCA of the semantic differential test results for the various soundscapes using eight semantic attributes (Table 3). The Kaiser–Meyer–Olkin (KMO) measures of sampling adequacy in the audio-only and audio-visual environments were 0.72 and 0.70, respectively, indicating optimal levels. Bartlett's test of sphericity produced clear results ($p < 0.01$) for each EE. Based on these results, the soundscapes were classified into two components. For the audio-only environment, Component 1 exhibited 33.67% explanatory power and high loading with "pleasant," "unpleasant," "calm," and "chaotic," and was therefore labeled *pleasantness*. Component 2 exhibited 32.75% explanatory power and contained "eventful," "uneventful," "exciting," and "monotonous," and was therefore labeled *eventfulness*. Application of the *pleasantness* and *eventfulness* components to the audio-visual environment had explanatory powers of 30.99 and 36.94%, respectively. These classifications coincide with those used in previous studies [59,60,67].

Appendix C shows the correlations between each soundscape-component factor score and the sound-source perception rates in different EEs. We converted variables belonging to one factor into a one-factor score calculated via regression analysis [68]. In the audio-only environment, *pleasantness* exhibited negative correlation with "traffic" and positive correlations with "human," "bird," and "music," while *eventfulness* showed positive correlations with "human," "traffic," and "music." In the audio-visual environment, *pleasantness* showed clearer correlations with "traffic" and "bird" and a lower correlation with "human," while *eventfulness* had slightly higher correlations with "traffic" and "human." In particular, the correlations between the perception of music and soundscape components when visual information was added was found to be statistically insignificant. An inspection of Fig. 1 and Table 1 reveals that this lack of correlation could arise from the limited number of ELs in which music was provided; by contrast, the perception rates for other sound sources increased with added visual information. This suggest that, even within a given acoustic environment, soundscape perception depends on visual information input.

3.2.2. Landscape principal components

We performed PCA on 12 landscape-related semantic attributes extracted in a common manner from the semantic differential test results (Table 3). The KMO measures of sampling adequacy in the visual-only and audio-visual environments were 0.75 and 0.72, respectively, indicating optimal levels. The results of Bartlett's test of sphericity were also clear for each EE ($p < 0.01$). These results indicate that the landscape components were classified differently depending on the EE. For the visual-only environment, Components 1–3, expressed as *overall quality* (including "comfortable" and "harmonious"), *spatial impression* (including "open" and "wide"), and *regularity* (including "orderly"), respectively, were derived. The explanatory powers of these landscape components were 27.66, 19.02, and 15.69%, respectively. For the audio-visual environment, Components 1–4, expressed as *overall quality, spatial impression, regularity*, and *naturalness* (a new component), respectively, were derived.

With respect to the soundscape components, Appendix D lists the correlations between each landscape-component factor score and visual-element perceptions in different EEs. In the visual-only environment, *overall quality* showed high negative correlations with "vehicle" and "road," and high positive correlation with "vegetation." *Regularity* had weak correlations with "vehicle," "building," and "sky." *Spatial impression* showed weak positive correlations with general visual elements but strong positive correlations with open areas in particular.

By contrast, in the audio-visual environment "vegetation" showed no correlation with *overall quality* but a positive correlation with *naturalness*, which in turn had relatively high negative correlations with artificial elements ("vehicle," "building," and "road"). Unlike under the visual environment, *regularity* had correlations with "road," "open space," and "vegetation," suggesting that visual-element perception can depend on the presence/absence of sound, even when the landscape components remain unchanged. *Spatial impression* had correlations with "open area" and "sky" alone but positive correlations with the visual elements in general. Thus, landscape perception clearly varies with the provided audio information provision even when the visual environment remains unchanged.

3.3. Overall satisfaction

We investigated overall subject satisfaction under different audio and visual information-provision scenarios and ELs. The highest satisfaction was generally reported in the visual-only environment (mean: 3.51, SD: 0.77), followed by the audio-visual (mean: 3.32, SD: 0.85) and audio-only environments (mean: 2.72, SD: 0.95) (Fig. 4). The differences in the averages among the three EEs were statistically significant ($F(2,538) = 104.132$, $p < 0.01$, $\eta_p^2 = 0.28$). Paired-sample t-testing was conducted to determine whether the mean difference in the subjects' satisfaction was significant. The difference in satisfaction between the cases in which only audio or visual information was provided and in which both were provided at the same time was statistically significant ($t(269) = -10.31$, $p < 0.01$, $\eta_p^2 = 0.63$; $t(269) = 3.81$, $p < 0.01$, $\eta_p^2 = 0.23$, respectively). Based on the change in the mean value of satisfaction in each environment, the relative contributions of audio and visual information were found to be approximately 24 and 76%, respectively, indicating the dominant influence of visual information and demonstrating that the overall preferences for urban spaces were more dependent on visual information.

4. Discussion

4.1. Audio-visual interaction for soundscape variables and landscape index perception

To examine the interactions between the perceptions of soundscape and landscape in an urban space, we analyzed the perceived sounds, visual elements, and PCA-derived correlations between soundscape and landscape (Table 4). We then compared any correlations exceeding the minimal effect (0.2) to the Cohen convention [69] for Pearson correlation. Among the sound sources, "traffic" showed strong positive correlations with "traffic" and "vehicle," but negative correlations with the *overall quality*, *regularity*, and *naturalness* of landscape components. "Vehicle" and "road" showed high negative correlations (>–0.5) with the *pleasantness* soundscape component. These results suggest that, as the presence of vehicles and roads increases, traffic perception rises, creating a negative effect in terms of urban visual satisfaction.

Vehicles and roads also had a clear negative effect on urban acoustic environment comfort from a soundscape perspective. Human sounds showed a strong positive correlation with people. Among nature-related sounds, "bird" showed a positive correlation with "vegetation" and negative correlations with "vehicle" and "road," coinciding with the results of previous studies [46,70]. Among the sound sources, "bird" showed the highest positive correlations with *overall quality* and *naturalness*, suggesting that bird sounds positively affect the overall impression of an urban landscape and can increase positive reactions to nature in urban green spaces. "Wind" was better perceived as the proportion of "sky" increased but showed no clear correlation with "music." In terms of correlations between soundscape and landscape components, *pleasantness* showed positive correlations with the overall landscape impression, the city's visual organization effectiveness, and the number of natural elements, while *eventfulness* was negatively correlated with *regularity*, suggesting that more acoustic events occur in the spaces of a more visually complex city.

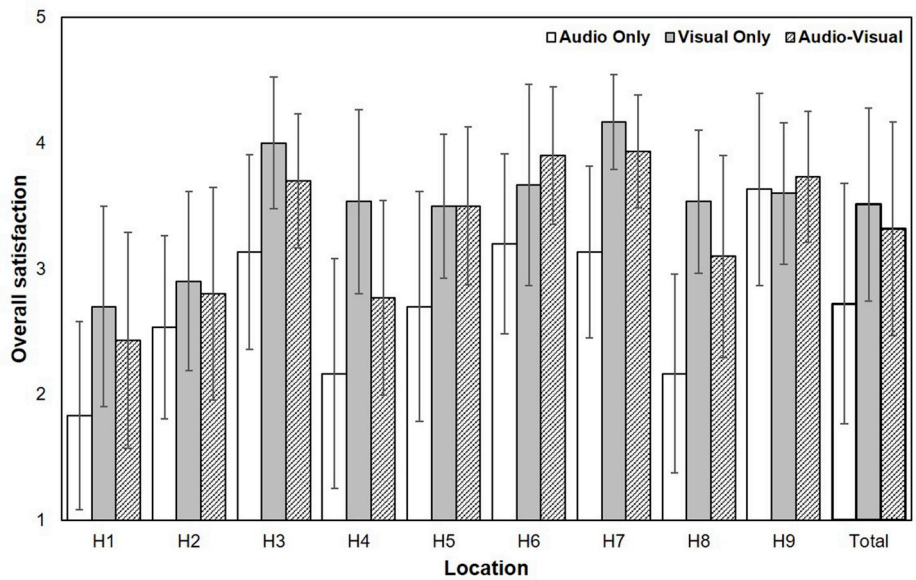

Fig. 4. Overall satisfaction for each location under different experimental set-ups.

Table 4
Correlation coefficients between perceived soundscape and landscape indices (*p < 0.05, **p < 0.01).

		Sound sources					Soundscape components	
		Traffic	Human	Bird	Wind	Music	Pleasantness	Eventfulness
Visual elements	**Vehicle**	0.72**	0.07	−0.22**	0.05	−0.12	−0.56**	0.05
	Building	0.29**	0.16**	−0.04	0.13*	−0.04	−0.21**	0.08
	Road	0.63**	0.06	−0.19**	0.06	−0.04	−0.53**	−0.03
	Open	−0.04	0.04	0.18**	0.01	−0.06	0.02	−0.12
	Vegetation	−0.07	0.05	0.25**	0.15*	0.17**	0.08	−0.11
	People	0.09	0.43**	0.04	0.18**	−0.01	−0.01	0.05
	Sky	0.01	0.22**	−0.05	0.22**	−0.14*	−0.01	0.24**
Landscape components	**Overall quality**	−0.28**	0.05	0.30**	0.11	0.01	0.50**	0.17**
	Regularity	−0.22**	0.01	0.17**	0.12	0.03	0.30**	−0.24**
	Spatial impression	0.09	0.10	−0.01	0.12	−0.03	−0.06	0.11
	Naturalness	−0.32**	−0.09	0.23**	0.08	0.12**	0.26**	−0.11

4.2. Relationship between overall satisfaction and audio-visual interaction

For each EE, we performed an MLR analysis to examine the contribution of audio-visual information to overall satisfaction. In the analysis, the perceived sound and visual elements were defined as independent variables and the potential confounding effect was taken into account by setting the evaluation environment order and subjects' genders as dummy variables. The standardized regression coefficients (β) obtained from the analysis are listed in Table 5. All the regression

Table 5
Standardized regression coefficients (β) obtained from multiple linear regression (MLR) analysis of overall satisfaction using sound and visual elements and soundscape and landscape components under different experimental set-ups (*p < 0.05, **p < 0.01).

A: Sound and visual elements

Effect	Environment	R^2	Traffic	Human	Bird	Wind	Music	Vehicle	Building	Road	Open	Vegetation	People	Sky
Visual	Audio Only	0.14	−0.27**	0.12*	0.12*	−0.14*	0.02	–	–	–	–	–	–	–
	Audio-visual	0.25	−0.29**	0.03	0.31**	−0.01	0.07	–	–	–	–	–	–	–
Audio	Visual Only	0.15	–	–	–	–	–	−0.28**	0.06	−0.05	0.17*	0.12	−0.08	0.05
	Audio-visual	0.21	–	–	–	–	–	−0.36**	−0.02	−0.10	0.07	0.11	0.02	−0.06
Interaction	Audio-visual	0.31	−0.02	0.05	0.29**	0.02	0.04	−0.30**	−0.03	−0.07	0.03	0.04	−0.01	−0.04

B: Soundscape and landscape components

Effect	Environment	R^2	Pleasantness	Eventfulness	Overall quality	Regularity	Spatial impression	Naturalness
Visual	Audio Only	0.45	0.66**	0.12**	–	–	–	–
	Audio-visual	0.32	0.56**	−0.05**	–	–	–	–
Audio	Visual Only	0.49	–	–	0.62**	0.28**	0.16**	–
	Audio-visual	0.49	–	–	0.55**	0.29**	0.05	0.32**
Interaction	Audio-visual	0.51	0.21**	−0.05	0.45**	0.22**	0.06	0.25**

models for overall satisfaction derived for the respective environments were statistically significant ($p < 0.01$). The order of gender and evaluation environment was not statistically significant, confirming that satisfaction did not vary by gender or order of environment evaluation. A model using perceived sound sources only as independent variables was able to predict or explain overall satisfaction with 14 and 25% explanatory power (R^2) for audio-only and audio-visual environments, respectively, indicating a slight increase in satisfaction with the provision of visual information. "Traffic" had a negative effect, with β values of −0.27 and −0.29 in the respective environments. Interestingly, both the "human" and "wind" contributions were beneficial, although weak, in the audio-only environment but insignificant when visual information was provided. The "bird" contribution was high (0.31). These results suggest that, in an audio-only environment, the subjects' overall satisfaction with urban space was determined based on a comprehensive assessment of various sound sources. With the addition of visual information, the subjects began to focus only on specific sounds, indicating that satisfaction was determined through the perception of a reduced number of sound types relative to the audio-only environment.

A model using perceived visual elements only as independent variables had 15 and 21% explanatory powers for the visual-only and audio-visual environments, respectively. Similar to the sound source results, "vehicle" had the highest β values: −0.28 and −0.36 in the respective environments. "Open space" had a significant contribution in the visual environment but no significant contribution when audio information was added. By contrast, the "vehicle" contribution increased with the provision of audio information. These results suggest that, even when both visual and audio information are provided, the subjects determined their satisfaction based on a focus on more conspicuous visual elements.

A model with both sound sources and visual elements as independent variables had a 31% explanatory power, which was higher than those of the prediction models derived using only one sensory information type. "Bird" and "vehicle" had the highest auditory and visual contributions, respectively. These results suggest that urban user satisfaction can be improved acoustically by enabling an increased perception of bird sounds and through urban planning in which the visible flow of vehicles is reduced. However, the overall explanatory power of the proposed regression model is low, probably because factors other than audio or visual information were insufficiently reflected in the urban-space satisfaction determination—a limitation of this study. In future studies, higher explanatory power could be obtained by considering additional physical indicators for urban acoustic and visual environments.

To examine the contributions of soundscape and landscape components to overall satisfaction, we performed an MLR analysis for each EE (Table 5B) using factors derived via a semantic differential test as independent variables. The resulting prediction regression models were statistically significant ($p < 0.01$), although the prediction models with soundscape components had explanatory powers of only 45 and 32% for the audio-only and audio-visual environments, respectively, with *pleasantness* making the largest contribution. Interestingly, the contribution of *eventfulness* varied significantly with environment, exhibiting positive and negative contributions to overall satisfaction in the audio-only and audio-visual environments, respectively. This suggests that, although the subjects considered feelings of activity or diversity indicated by sounds to be positive, negative perceptions affecting the overall satisfaction could develop depending on the urban visual composition, as suggested by the negative correlation between *regularity* and *eventfulness* in Table 4.

The prediction model with landscape components had only 49% explanatory power for both visual-only and audio-visual environments, with the overall quality making the largest positive contribution to *overall satisfaction*. The largest differences between the two EEs appeared for *spatial impression* and *naturalness*. In the visual-only environment, the sense of open space in the urban area significantly influenced the *overall satisfaction* evaluation but made no contribution to the audio-visual environment. The contribution to *naturalness* was high, at 0.32, suggesting that, in perceiving an urban landscape considering the soundscape elements, the subjects compared the naturalness and artificiality of the space rather than perceiving the sense of overall space.

A regression model using all soundscape and landscape components as independent variables had 51% explanatory power. The factor with the largest contribution to overall satisfaction was *overall quality* (β: 0.45), followed by *naturalness* (β: 0.25), *regularity* (β: 0.22), and *pleasantness* (β: 0.21). In general, landscape elements made higher contributions than soundscape elements, suggesting that the subjects were more strongly affected by visual elements in evaluating their urban space satisfaction.

5. Conclusions

In this study, we used VR-simulated acoustic and visual environments to investigate the effects of audio-visual interactions on soundscape and landscape perception by humans with the goal of making urban spaces more satisfying. To enable an immersive and realistic soundscape evaluation within a limited laboratory environment, information on space and direction were provided through FOA reproduction, while realistic visual information was provided through an HMD. To examine the influence of individual audio and visual information sources on overall urban space satisfaction in human subjects and reveal relevant audio and visual interactions, a number of factors were subjectively evaluated, including perceived audio-visual elements, audio-visual semantic attributes, and satisfaction at different EEs.

The EE-dependent evaluation results for the perceived elements confirmed the influence of both audio and visual information. The sound sources (coded as "traffic," "human," "bird," and "wind") and visual elements ("vehicle," "building," "vegetation," "people," and "sky") were perceived differently depending on the audio and visual information provided. The influence of sound sources varied with location and represented a limitation on the results, indicating that further investigation at a more diverse set of locations is needed. Based on the results of a semantic differential evaluation, the soundscape components were classified in terms of *pleasantness* and *eventfulness* and the landscape components were classified in terms of *overall quality, regularity, spatial impression,* and *naturalness.* In particular, the *naturalness* component appeared only when audio information was provided, suggesting that audio information is critical to the perception of the nature-friendliness of an urban space. The overall satisfaction evaluation results revealed visual and audio information contributions of 76% and 24%, respectively, indicating the importance of the visual aspect in urban audio-visual environment design.

Based on the audio-visual elements and soundscape and landscape components, a regression model was proposed. The satisfaction prediction capability of the model via a combination of soundscape and landscape components was found to be sufficient and effective for use in urban space design and enhancement. A model using only audio-visual elements had a rather low explanatory power of 31% but revealed the significant influence of bird sounds and vehicle visual elements—two major urban design factors—on satisfaction. A model combining soundscape and landscape components had a high explanatory power of 51%. It was found that *naturalness* and *overall quality* both improved through the addition of visually well-organized and appropriate vegetation elements, while *pleasantness* improved through the provision of an acoustically comfortable space, contributing effectively to the overall satisfaction with urban space.

The study was limited in that the target urban spaces were all located in Seoul and all the subjects were in their twenties and, therefore, the results might not be applicable to all cities and age groups. To generalize the study results to a more diverse urban space and demographic, evaluation over a wide range of age groups in urban spaces with various functionalities will be essential. In addition, broader, internationally focused research should be performed to enable a comparison between

results for Eastern and Western countries with different urban densities and morphologies. Despite these limitations, the audio-visual interactions of soundscape and landscape uncovered in this study and the satisfaction models derived based on the interactions should provide effective and concrete guidelines to improve satisfaction levels in newly designed urban spaces.

Declarations of competing interest

There are no conflicts of interest to declare.

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Appendix A. Supplementary data

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